**Learning Objective 7.3:** I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **A coax with a thick shield.** A coaxial line has an inner conductor of radius $a = 1$ mm and a shield that fills $b = 4$ mm $< \rho < c = 5$ mm. It carries $I = 2$ A at DC, out along the inner conductor and back along the shield, with the current uniform over each conductor's cross section.

   (a) Use Ampere's law to find $H_\phi$ in all four regions.

   The path is a circle of radius $\rho$, so $2\pi\rho\, H_\phi = I_{\text{encl}}$ in each region.

$$\begin{aligned}
\rho < a: &\quad I_{\text{encl}} = I\frac{\rho^2}{a^2}, &\quad H_\phi &= \frac{I\rho}{2\pi a^2} \\
a < \rho < b: &\quad I_{\text{encl}} = I, &\quad H_\phi &= \frac{I}{2\pi\rho} \\
b < \rho < c: &\quad I_{\text{encl}} = I - I\frac{\rho^2 - b^2}{c^2 - b^2} = I\frac{c^2 - \rho^2}{c^2 - b^2}, &\quad H_\phi &= \frac{I}{2\pi\rho}\,\frac{c^2 - \rho^2}{c^2 - b^2} \\
\rho > c: &\quad I_{\text{encl}} = 0, &\quad H_\phi &= 0
\end{aligned}$$

   (b) Find $H_\phi$ at $\rho = 0.5$ mm, 2 mm, 4.5 mm, and 6 mm.

$$H(0.5) = \frac{2 \times 0.5 \times 10^{-3}}{2\pi \times (10^{-3})^2} = 159 \text{ A/m}$$

$$H(2) = \frac{2}{2\pi \times 0.002} = 159 \text{ A/m}$$

$$H(4.5) = \frac{2}{2\pi \times 0.0045} \times \frac{25 - 20.25}{25 - 16} = 37.3 \text{ A/m}$$

$$H(6) = 0$$

| | |
|---|---|
| 0.5 mm: | 159 A/m |
| 2 mm: | 159 A/m |
| 4.5 mm: | 37.3 A/m |
| 6 mm: | 0 |

2. **A finite solenoid.** A coil of $N = 1000$ turns is wound over $\ell = 20$ cm on a form of radius $a = 2$ cm and carries $I = 0.5$ A, in air.

   (a) Find $H$ and $B$ inside, treating the coil as a long solenoid.

$$n = \frac{1000}{0.20} = 5000\ \text{m}^{-1}, \qquad H = nI = 2500\ \text{A/m}$$
$$B = \mu_0 nI = 4\pi \times 10^{-7} \times 2500 = 3.14\ \text{mT}$$

   $H$ = 2500 A/m

   $B$ = 3.14 mT

   (b) Use the finite-solenoid formula to find $B$ at the center and at one end of the coil.

$$B(0) = \mu_0 nI\ \frac{\ell}{\sqrt{\ell^2 + 4a^2}}$$
$$= 3.14 \times \frac{0.20}{\sqrt{0.04 + 0.0016}} = 3.08\ \text{mT}$$
$$B(\ell/2) = \frac{\mu_0 nI}{2}\ \frac{\ell}{\sqrt{\ell^2 + a^2}}$$
$$= 1.57 \times \frac{0.20}{\sqrt{0.04 + 0.0004}} = 1.56\ \text{mT}$$

   center: 3.08 mT

   end: 1.56 mT

   (c) How long must a coil of radius 2 cm be for its center field to come within 1% of $nI$?

$$\frac{\ell}{\sqrt{\ell^2 + 4a^2}} \geqslant 0.99 \ \Rightarrow\ \ell^2\,(1 - 0.9801) \geqslant 0.9801 \times 4a^2$$
$$\ell \geqslant a\sqrt{\frac{3.920}{0.0199}} = 14.0\,a = 28\ \text{cm}$$

   $\ell$ = 28 cm

3. **A toroid.** A toroid of $N = 400$ turns has a rectangular cross section with inner radius $r_1 = 2$ cm, outer radius $r_2 = 3$ cm, and height $h = 1$ cm. It carries $I = 0.25$ A and has an air core.

   (a) Find $H$ at the inner and outer edges of the core.

$$H(r_1) = \frac{NI}{2\pi r_1} = \frac{100}{2\pi \times 0.02} = 796 \text{ A/m}$$
$$H(r_2) = \frac{100}{2\pi \times 0.03} = 531 \text{ A/m}$$

$H(r_1)$ = 796 A/m

$H(r_2)$ = 531 A/m

   (b) Derive the magnetic flux $\Phi$ through one turn, the rectangle $r_1 < \rho < r_2, 0 < z < h$, and evaluate it.

$$\Phi = \int_0^h \int_{r_1}^{r_2} \frac{\mu_0 NI}{2\pi\rho}\, d\rho\, dz = \frac{\mu_0 NIh}{2\pi} \ln \frac{r_2}{r_1}$$
$$= 2 \times 10^{-7} \times 100 \times 0.01 \times \ln 1.5 = 8.11 \times 10^{-8} \text{ Wb}$$

   (c) The current path passes through the core $N$ times, so the flux it links is $N\Phi$. Find the inductance $L = N\Phi/I$.

$$L = \frac{400 \times 8.11 \times 10^{-8}}{0.25} = 1.30 \times 10^{-4} \text{ H} = 130\ \mu\text{H}$$

$L$ = 130 $\mu$H

4. **RG-59.** RG-59 has $a = 0.29$ mm, $b = 1.85$ mm, and solid polyethylene with $\epsilon_r = 2.25$. Lesson 25's practice set found $C' = 67.5$ pF/m.

   (a) Find $L'$, $Z_0$, and $v_p$.

$$L' = \frac{\mu_0}{2\pi} \ln \frac{b}{a} = 2 \times 10^{-7} \times \ln 6.379$$
$$= 2 \times 10^{-7} \times 1.853 = 371 \text{ nH/m}$$
$$Z_0 = \sqrt{\frac{3.71 \times 10^{-7}}{6.75 \times 10^{-11}}} = 74.1\ \Omega$$
$$v_p = \frac{1}{\sqrt{L'C'}} = \frac{c}{\sqrt{2.25}} = 2.00 \times 10^8 \text{ m/s}$$

$L'$ = 371 nH/m

$Z_0$ = 74.1 Ω

$v_p$ = $2.00 \times 10^8$ m/s

   (b) What ratio $b/a$ gives $Z_0 = 50\ \Omega$ with the same dielectric?

$$\eta = \sqrt{\mu/\epsilon} = 377/\sqrt{2.25} = 251\ \Omega$$
$$Z_0 = \frac{\eta}{2\pi} \ln \frac{b}{a} = \frac{251}{2\pi} \ln \frac{b}{a} = 40 \ln \frac{b}{a}$$
$$\ln \frac{b}{a} = \frac{50}{40} = 1.25, \qquad \frac{b}{a} = e^{1.25} = 3.49$$

$b/a$ = 3.49

**Documentation:** _____